Statistical Thinking for Data Science

Week 11: Regression and prediction

Fitting a line, adjusting for something else, and knowing which job you are doing

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Lecture notes for students of Statistical Thinking for Data Science.

Week 11: Regression and prediction

Learning Objectives

By the end of this week you should be able to:

- fit a straight line by least squares and interpret both coefficients in the units of the problem;
- quantify the uncertainty in a slope and test it;
- interpret R^2 and the residual standard deviation, and say what each misses;
- read a residual plot and name the assumption each pattern violates;
- explain what "adjusting for" a second variable means, and what collinearity does to the standard errors;
- distinguish prediction from explanation, and evaluate a predictive model on data it has not seen.

11.1 The model

Week 5 measured how two variables move together. Regression goes further and writes down a model:

$$Y = \beta_0 + \beta_1 x + \varepsilon, \qquad \varepsilon \sim (0, \sigma^2) \text{ independent.}$$

Read it as a statement about averages: for a given x, the response varies around a mean of $\beta_0 + \beta_1 x$, with scatter σ. The line is a claim about the *mean* of Y, never about any individual.

Least squares picks the line minimising $\sum (y_i - \hat{y}_i)^2$, giving

$$\hat{\beta}_1 = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2}, \qquad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}.$$

The fitted line always passes through $(\bar{x}, \bar{y})$, which is a useful check and a useful intuition: the line pivots about the centre of the data.

Example 1: study hours and exam mark

Ten students report weekly study hours and sit the same exam.

Hours x	2	3	4	5	5	6	7	8	9	11
Mark y	44	45	58	52	63	58	72	65	78	80

Solution

$\bar{x} = 6.0$, $\bar{y} = 61.5$, and the sums give $\hat{\beta}_1 = 4.20$ and $\hat{\beta}_0 = 36.30$:

$$\widehat{\text{mark}} = 36.3 + 4.2 \times \text{hours}.$$

Slope: each extra hour of study a week is associated with 4.2 more marks, on average, in this sample.
Intercept: 36.3 is the fitted mean mark at zero hours. Nobody in the data studied zero hours, so this is an extrapolation and should be treated as a number that positions the line, not as a claim about non-studying students.
$r = 0.928$ and $R^2 = 0.862$. The residual standard deviation is $s_e = 4.97$ marks: even knowing study hours, individual students scatter about five marks either side of the line.

11.2 Is the slope real?

The slope is an estimate, so it has a standard error:

$$\mathrm{SE}(\hat{\beta}_1) = \frac{s_e}{\sqrt{\sum (x_i - \bar{x})^2}}.$$

For the example, SE $= 0.594$, so

$$t = \frac{4.20}{0.594} = 7.07 \quad \text{on } n - 2 = 8 \text{ df}, \qquad p = 0.0001,$$

and a 95% interval of $4.20 \pm 2.306(0.594) = (2.83,\ 5.57)$ marks per hour.

Report the interval. "Between about 3 and 5.5 marks per hour" is what a reader can use; "$p < 0.001$" only says the slope is not zero, which was never in doubt.

Notice the denominator: the more spread out your x values, the smaller the standard error. If you want a precise slope, measure at a wide range of x, not many times at the same x. This is the design lesson hidden inside the formula.

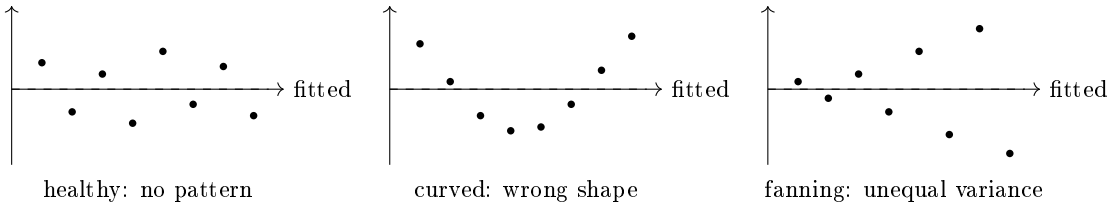
11.3 R^2, and what it does not tell you

R^2 is the share of the variance of y explained by the model: 0.862 here. Four cautions:

- It says nothing about whether the straight line is the right shape. A curved relationship can produce a high R^2 and systematically wrong predictions.
- It rises whenever you add a predictor, useful or not, so it cannot be used to choose between models of different size.
- A high R^2 in a narrow range of x can collapse outside it.
- For prediction, s_e is usually the more honest number: here, about five marks of irreducible scatter.

11.4 Checking the model: residuals

A residual is $e_i = y_i - \hat{y}_i$. Plot the residuals against the fitted values and look for structure. There should be none.



healthy: no pattern | curved: wrong shape | fanning: unequal variance

Pattern	What it means
Curve	the relationship is not linear: transform x, or fit a curve
Fanning out	the variance grows with the mean: transform y, often by logs
A few far-out points	influential observations: check them, and report the fit with and without
Drift over time or over row order	the observations are not independent

11.5 Extrapolation

The fitted line for our ten students predicts $36.3 + 4.2(20) = 120$ marks for twenty hours of study. Marks stop at 100, so the model is plainly wrong there. Nothing in the arithmetic warned us: least squares will extend a line as far as you ask.

The rule

A regression describes the range of x you observed, here 2 to 11 hours. Outside it you are assuming the pattern continues, which is a scientific claim about the world, not a result of the fit. State it when you make it, or do not make it.

11.6 More than one predictor, and what adjustment means

With two predictors,

$$Y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \varepsilon,$$

and β_1 now means: the expected change in Y for a one unit change in x_1 *with* x_2 *held fixed*. That phrase is the whole of "adjusting for". It is also where Week 5's confounding is dealt with: if the confounder is measured, you can put it in the model.

Example 2: adjusting for prior ability

Add each student's prior-year average to the model.

Solution

Fitted: $\widehat{\text{mark}} = 40.5 + 4.80\,\text{hours} - 0.13\,\text{prior}$, with standard errors 1.68 for hours and 0.33 for prior. R^2 rises only from 0.862 to 0.865.

So prior ability adds essentially nothing once hours are known, and its coefficient is swamped by its own standard error. More striking is what happened to hours: the estimate moved from 4.20 to 4.80, but its standard error nearly tripled, from 0.59 to 1.68, so the interval is now roughly 1 to 9 marks per hour.

Why

Hours and prior average correlate at 0.93 in this class: hard-working students were already doing well. When two predictors carry nearly the same information, the data cannot say which deserves the credit, and the model expresses that honestly by widening both intervals. This is **collinearity**. The variance inflation factor here is about 7, meaning the variance of the hours coefficient is seven times what it would be with uncorrelated predictors.

What to do

Not "drop variables until the p-values look good". Decide what the model is for. To *predict* marks, keep whichever set predicts best on new data and do not interpret the coefficients. To *explain* the effect of studying, you need a design that separates studying from prior ability, which observational data from one class cannot do.

11.7 Prediction and explanation are different jobs

	Prediction	**Explanation**
Question	what will Y be?	what happens to Y if we change x?
Good model	accurate on new data	unbiased for the effect of interest
Correlated predictors	no problem, keep them	a serious problem
Extra variables	include if they help	include only with a reason
Judged by	error on a held-out set	design, and control of confounding
Causal claim	not needed	required

A model can be excellent at one job and useless at the other. The number of ambulances at a scene predicts fatalities very well and would be a disastrous basis for policy.

11.7.1 Honest evaluation: hold data out

A model's error on the data used to fit it is optimistic, and the more flexible the model, the more optimistic. Split the data, fit on one part, measure error on the other; with little data, use cross validation, where each tenth of the data takes a turn as the test set. Report the error on data the model has not seen, and report the units: "typical error 5.2 marks" means something, "$R^2 = 0.86$" less so.

11.8 Reporting a regression

State the model, the coefficients with intervals and units, the sample size, the residual standard deviation, the range of x the model covers, and what you checked. For example:

> Marks rose by 4.2 per extra weekly study hour (95% CI 2.8 to 5.6; $n = 10$, $R^2 = 0.86$, residual sd 5.0 marks, hours ranging 2 to 11). Residuals showed no curvature or fanning. This is an observational association: students who study more also entered with higher averages, and the two cannot be separated in this sample.

11.9 Doing it on a computer

R: `fit <- lm(mark ~ hours, data = d)`, then `summary(fit)`, `confint(fit)` and `plot(fit)` for the four diagnostic plots. Python: `statsmodels.formula.api.ols("mark ~ hours", d).fit()` and `.summary()`, or `sklearn` when the job is prediction and you want cross validation. Look at the diagnostic plots before you look at the p-values.

11.10 Glossary

Least squares	the line minimising the sum of squared residuals
Residual	observed minus fitted
R^2	the share of the variance of y explained by the model
Residual standard deviation	typical size of a residual, in the units of y
Adjustment	estimating one coefficient with the other predictors held fixed
Collinearity	predictors carrying nearly the same information, which inflates standard errors
Extrapolation	using the model outside the observed range of the predictors
Cross validation	measuring error on data the model did not see

11.11 Summary

Week 11 in six lines

1. A regression line is a statement about the mean of Y at a given x.
2. Report the slope with its interval and its units, not just its p-value.
3. R^2 does not check the shape; the residual plot does.
4. A line extended beyond your data is an assumption, not a result.
5. "Adjusting for" means holding fixed, and correlated predictors widen every interval.
6. Decide whether you are predicting or explaining, because the rules differ.

Exercises

1. Using the ten students in Example 1, verify $\hat{\beta}_1 = 4.2$ and $\hat{\beta}_0 = 36.3$ by hand, and confirm the line passes through $(6, 61.5)$.

2. Interpret the slope and the intercept in words a student would understand, and say which of the two is meaningful here.

3. Compute the 95% interval for the slope and state what it tells a reader that the p-value does not.

4. The model predicts 120 marks for twenty hours of study. Explain what went wrong, and state the rule you would write on the whiteboard.

5. Adding prior-year average changes the hours coefficient from 4.20 (SE 0.59) to 4.80 (SE 1.68), and R^2 from 0.862 to 0.865. Explain both changes, and say what you would do if your aim were to measure the effect of studying.

6. Sketch three residual plots: one healthy, one showing curvature, one showing unequal variance. For each unhealthy one, name a remedy.

7. A colleague reports $R^2 = 0.91$ for a model predicting next month's production, fitted and evaluated on the same 24 months. Describe the evaluation you would ask for instead, and why the 0.91 is optimistic.

8. Find a published regression in your field. Write down whether its authors were predicting or explaining, and whether their design supports the claims they make.

Answers to the odd-numbered exercises

1. $\sum(x-\bar{x})(y-\bar{y}) = 294.0$ and $\sum(x-\bar{x})^2 = 70.0$, so $\hat{\beta}_1 = 294/70 = 4.2$ and $\hat{\beta}_0 = 61.5 - 4.2(6) = 36.3$. At $x = 6$ the line gives $36.3 + 25.2 = 61.5 = \bar{y}$, as it must.

3. $4.2 \pm 2.306(0.594) = (2.83, 5.57)$ marks per hour. The p-value says only that the slope is not zero; the interval says the effect could be as small as three marks an hour or as large as five and a half, which is the difference between advice worth giving and advice worth shouting about.

5. The coefficient moved because the two predictors are correlated at 0.93, so the model redistributes credit between them; the standard error tripled because with such collinearity the data cannot separate their contributions, and R^2 barely moved because prior average adds almost no information beyond hours. To measure the effect of studying you need variation in study hours that is not tied to prior ability, which means a designed study, not this class.

7. Fitting and evaluating on the same 24 months measures how well the model memorised them. Ask for performance on months the model never saw: either a hold-out of the most recent months, or rolling origin cross validation for time series, reported as a typical error in tonnes rather than as R^2.

Before next week

Fit a straight line to any pair of variables you have, look at the residual plot before the coefficients, and bring both. Next week is the last: probability as belief, updating with evidence, and the analysis of the survey this class filled in.